Algorithmic Detection of Financial Economic Crime

Michael Attard, Louis de Bruijn

March 2026

ING Analytics

do your thing

Goal

(1) Present Machine Learning (ML) algorithms that can be used in the KYC, Fraud and Transaction Monitoring domain.

(2) With actual use-cases at ING

Who are we

Michael Attard

Studied Engineering Science @ University of Toronto & Astronomy @ Western University

Started in Data Science in 2017

Worked as Data Scientist in various different industries:

- Manufacturing
- Retail
- Finance

Working as a Data Scientist at ING since 2024 within Transaction Monitoring and Customer Due Diligence

Louis de Bruijn

Studied computational linguistics @ Rijksuniversiteit Groningen

Traineeship in Analytics at ING in 2020

Worked as Data Scientist for different departments

- Customer Dialogue Analytics
- Advanced Real Time Analytics
- Risk & Pricing Analytics

Now working as Data Scientist for Transaction Monitoring

Agenda

1. Obligations for Banks	(5min)	Michael
2. General insights on Machine Learning	(10min)	Michael
3. Precision/Recall Demo	(15min)	Louis / Both
4. Applied to Banking	(5min)	Louis
5. Generative AI in Banking	(5min)	Louis
6. CDD (RAM)	(10min)	Michael
7. Transaction Monitoring (SATM)	(10min)	Louis

Obligations for Banks

- **WWFT**
  - “Wet ter voorkoming van witwassen en financieren van terrorisme” (Money laundering and terrorist financing prevention law).

  - Money laundering
    - Money laundering involves making illegally acquired assets legal so that their illegal origin is no longer visible.

  - Terrorist financing
    - Using assets to facilitate terrorist activities.

- **Duties for banks**
  - Customer Due Diligence (CDD) / Know Your Customer (KYC).
  - Transaction Monitoring (TM) - Report unusual transactions.

Show of hands

Who has programmed in Python?

Who has used scikit-learn (sklearn) library?

Who knows why you’d use AUPRC on an imbalanced dataset?

General insights on Machine Learning

Traditional computer programs

Patterns are explicitly coded.

IF (Company is in *Virgin Islands* or *Cayman Islands*) AND (Amount > *100.000 USD*):
- Investigate further

ELSE:
- Do not investigate further

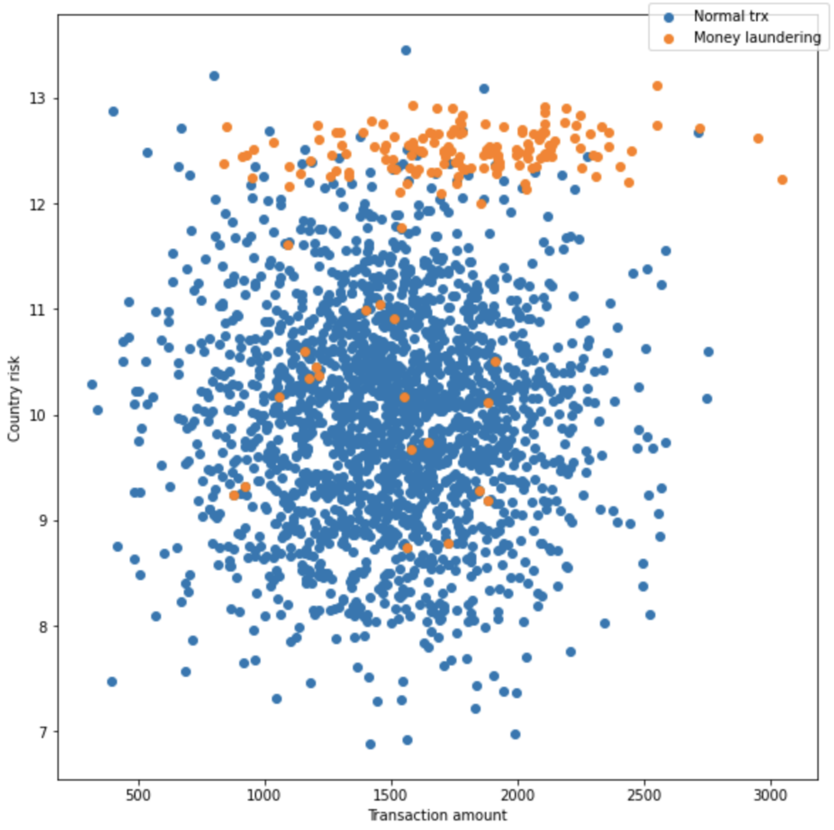
Deciding on which rules to implement requires a lot of data analysis and expert knowledge.

Very granular patterns are hard to find.

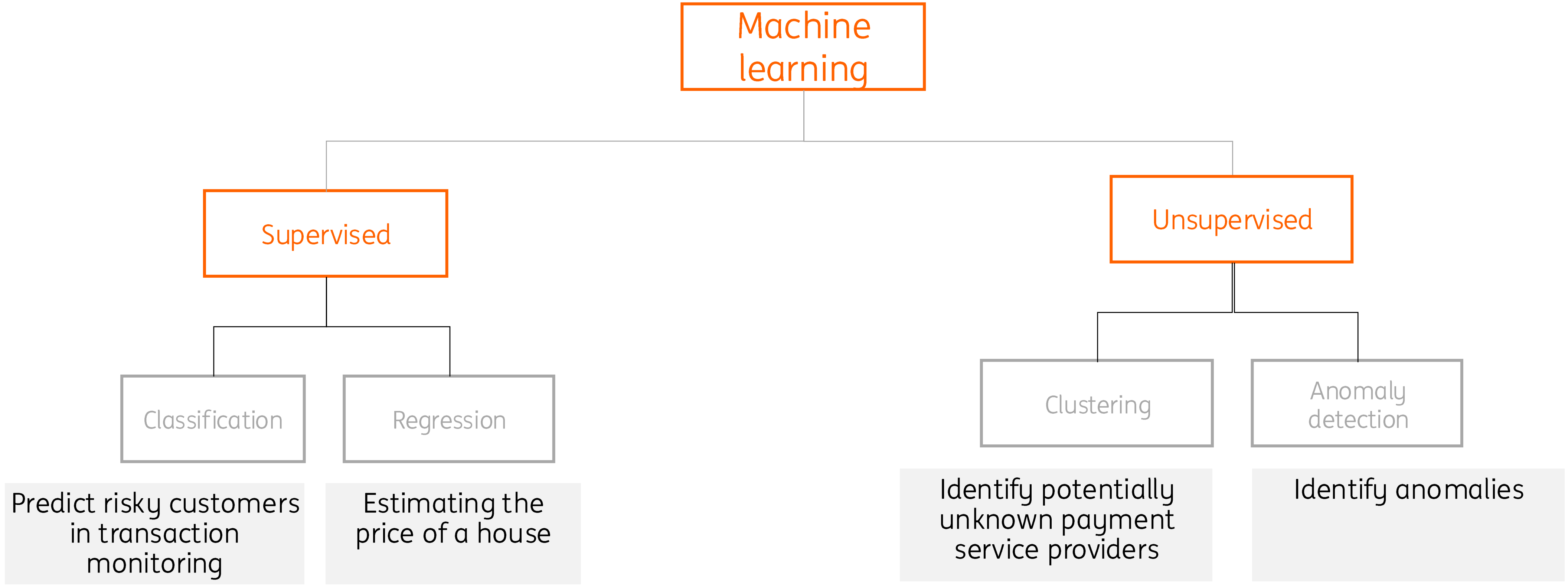
The more rules added, the more complexity to maintain this system.

Machine Learning

Patterns are derived from the data
Requires a lot of labels.

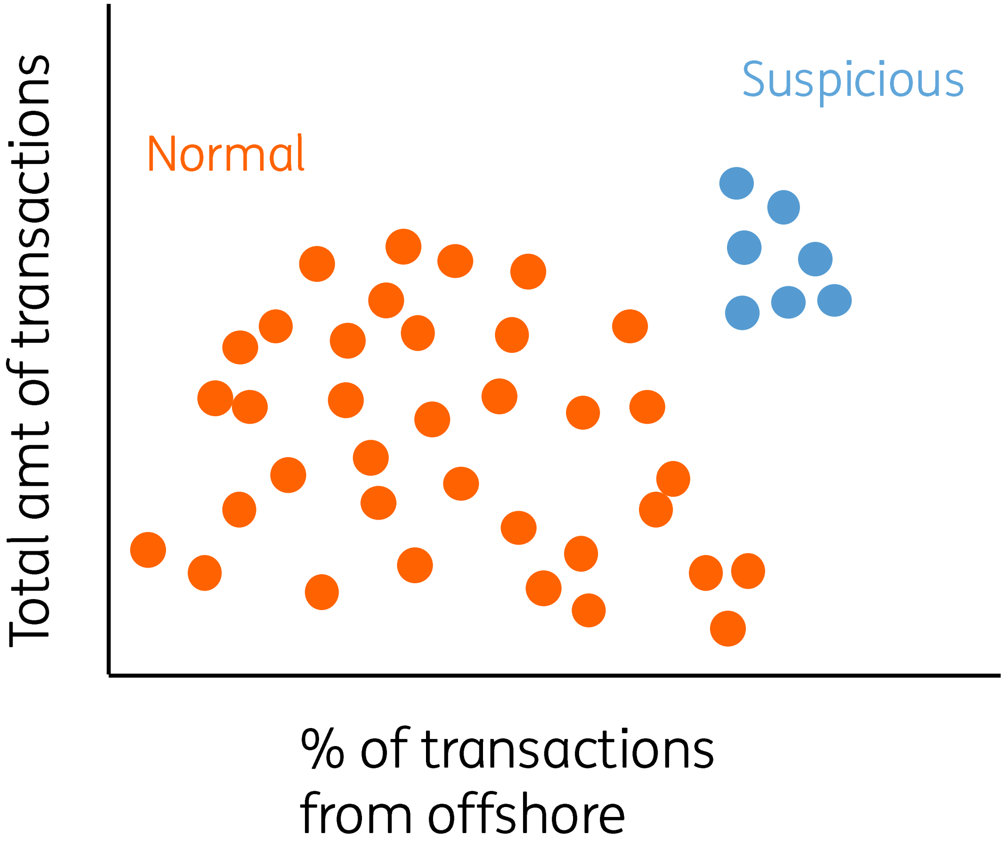


Different types of machine learning

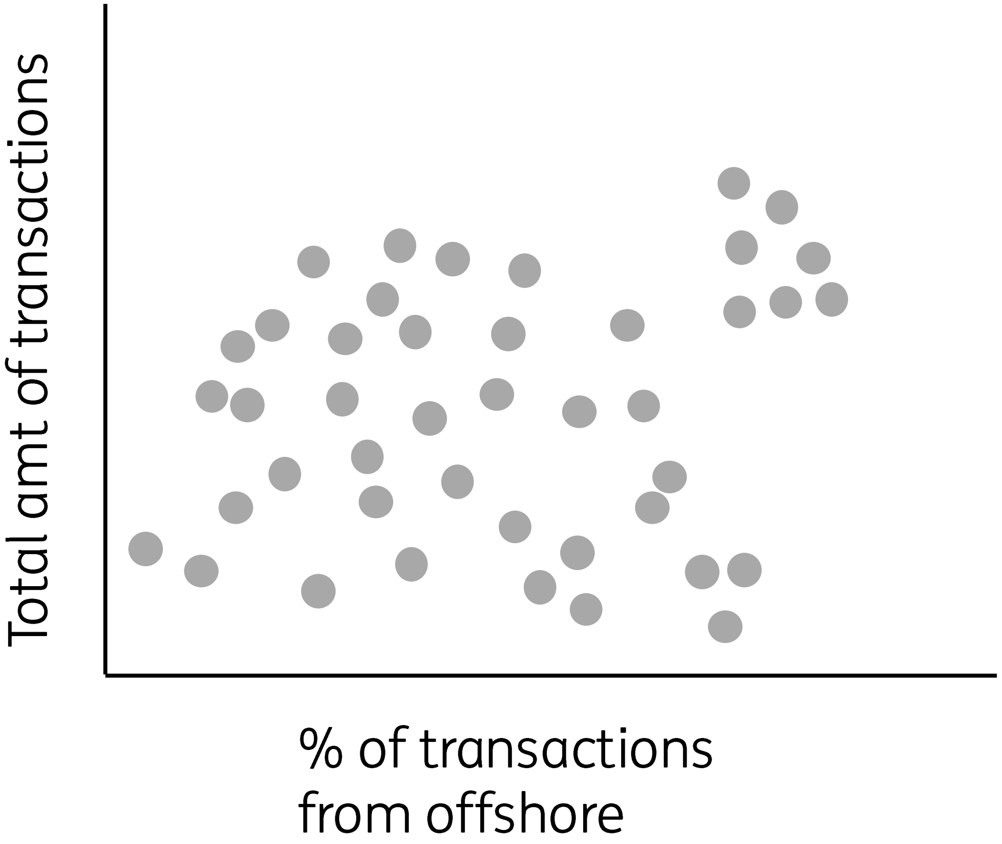


Supervised and unsupervised machine learning

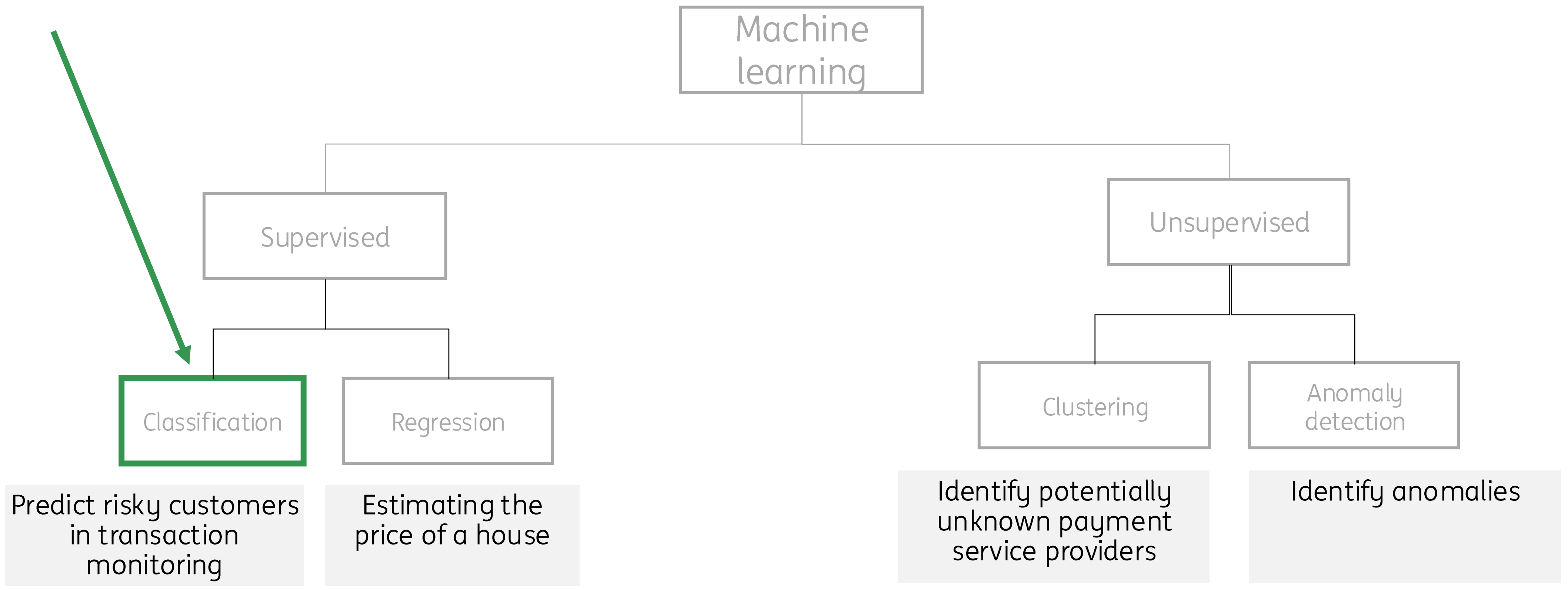
Supervised
With labeled data – find the best separation line



Unsupervised
No labels. Self-discover clusters.



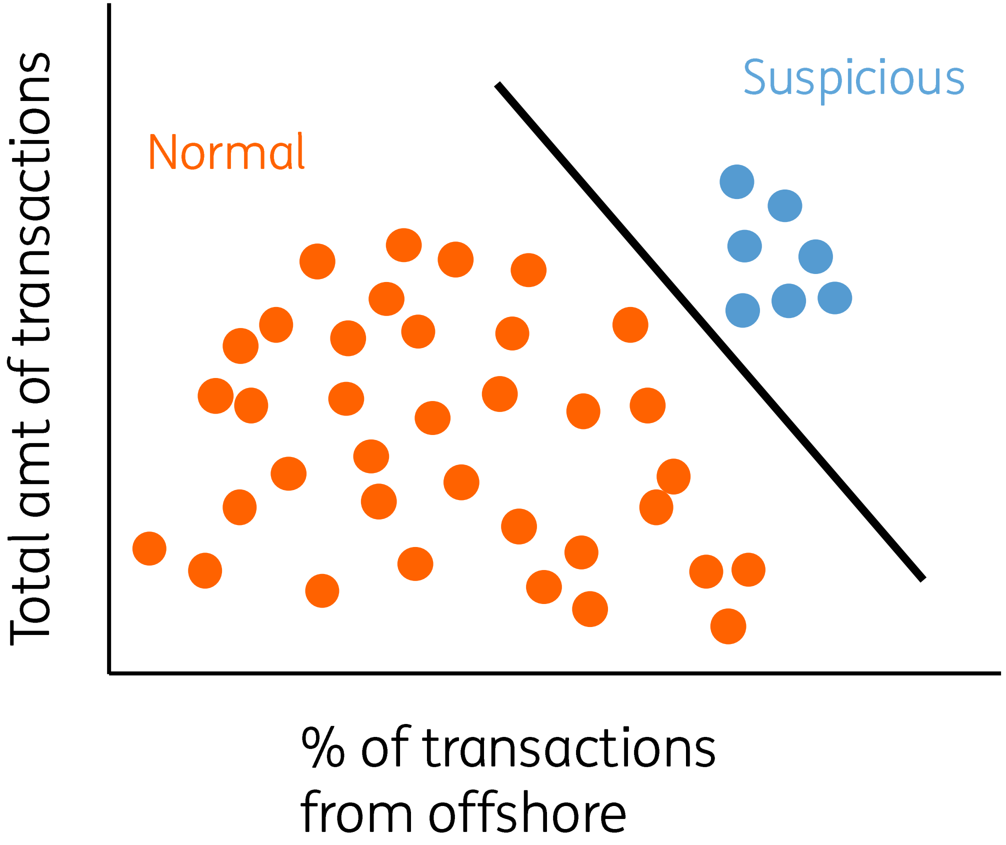
Different types of machine learning



Classification models

Find a way to divide up the data according to class

2 features
Total amt of tx
% of tx from offshore

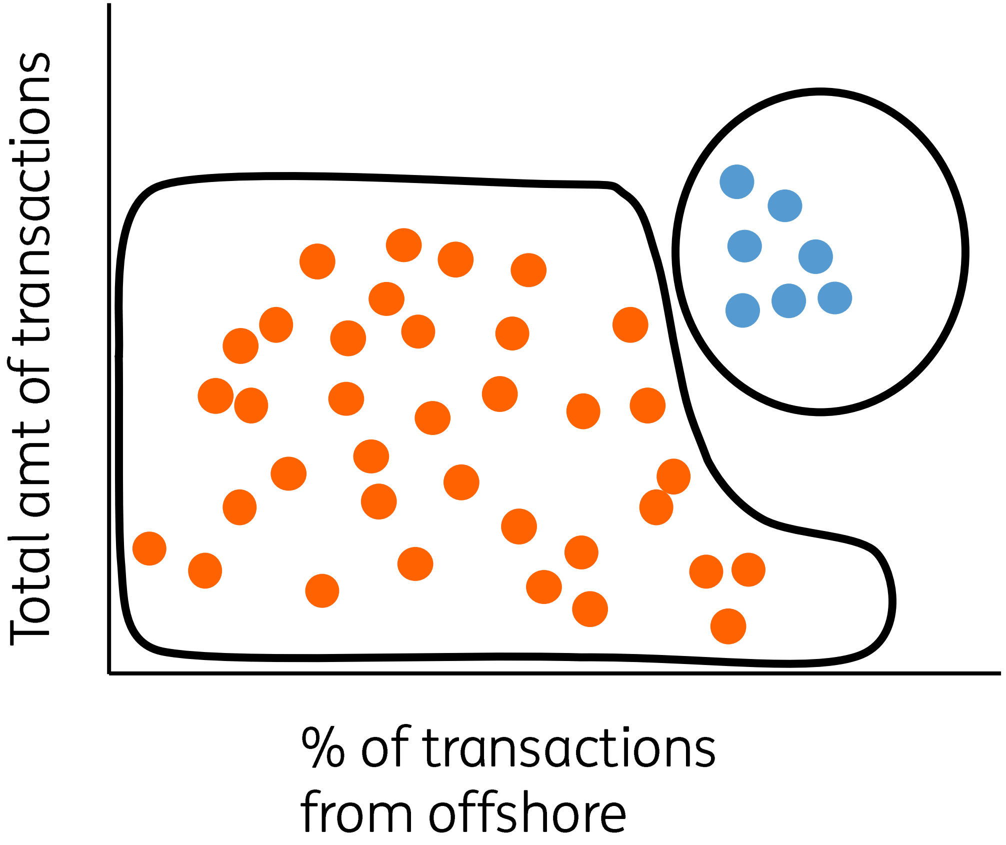


Find a line that best separates the two colors (classes)

Classification models

Find a way to divide up the data according to class

2 features
Total amt of tx
% of tx from offshore



Separation line doesn't have to be a line!

Different algorithms use different ways to draw the separation lines

Imbalanced data

Training a machine learning model requires a large amount of data

- If there are few risky companies, the model cannot distinguish which features indicate riskyness
- For example, from the first table, the algorithm could learn that “LTD” companies in the UK transferring large amounts are risky

Company	Country	Amount	Legal Entity	...	Is Risky?
ABC	NL	$4.000	BV	...	No
DEF	UK	$400.000	LTD	...	Yes
GHI	NL	$100	no	...	No
	...	...	...	...	...

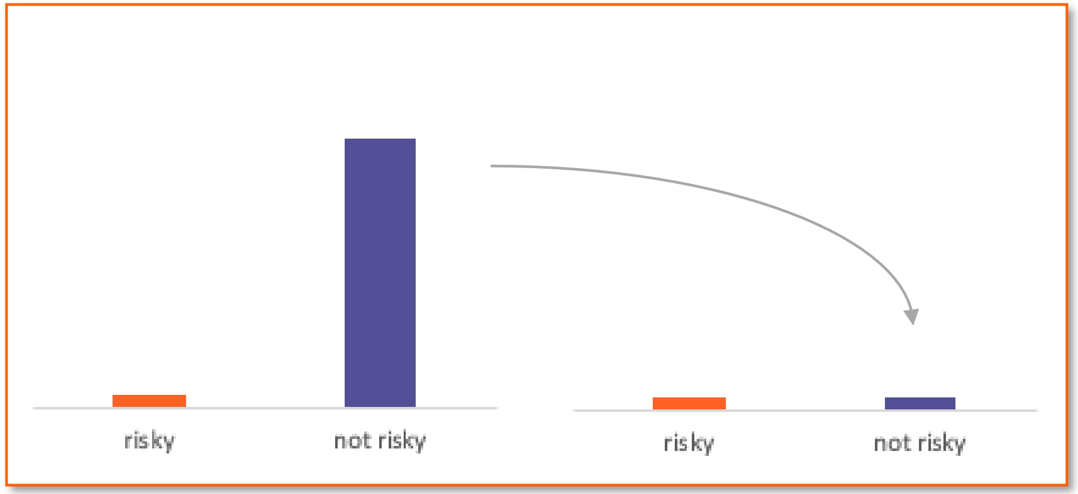
Imbalanced data

- But if more risky companies with different characteristics are added to the training data, the algorithm learns that the risk is more associated with “LTD” than the amount or country
- Only a small fraction of companies require expert investigation when it comes to economic crime
  - Therefore this data is called “imbalanced”
- A lot of amnually labelled data is required
- E.g. assume we need 1.000 “Yes” labels and only 5% of the companies are risky.
- We would need to manually label 20.000 companies to find 1.000 “Yes”

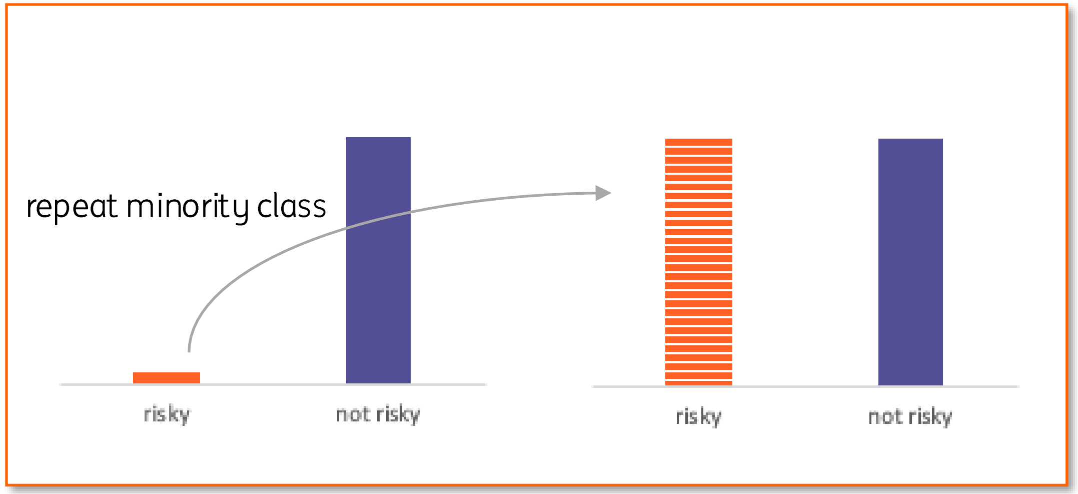
Company	Country	Amount	Legal Entity	...	Is Risky?
ABC	NL	$4.000	BV	...	**No**
DEF	UK	$400.000	LTD	...	**Yes**
GHI	NL	$100	no	...	**No**
LMN	US	$8.000	LTD	...	**Yes**
OPQ	NL	$20.000	LTD	...	**Yes**
	...	...	...	...	...

Solutions to deal with imbalanced data

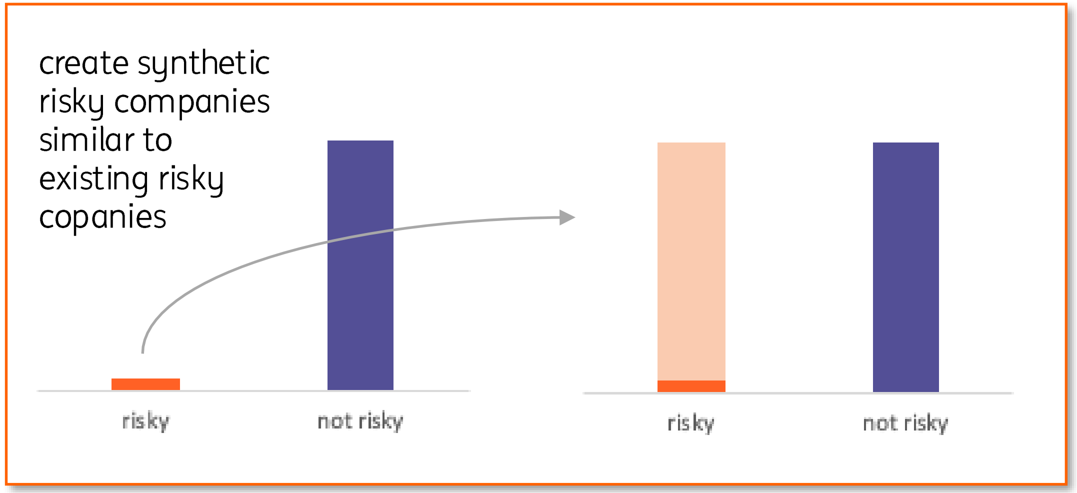
Undersampling majority class



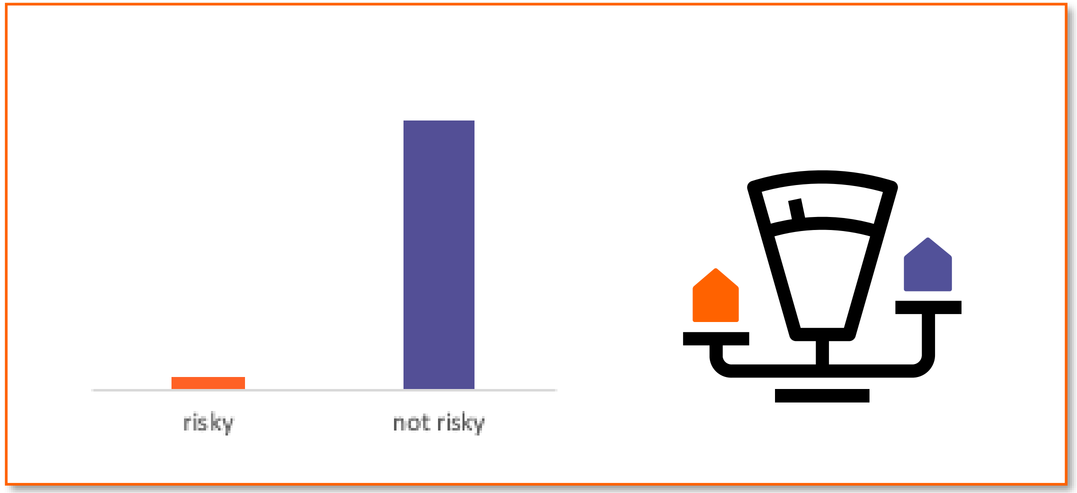
Oversampling minority class



Creating syntetic data points



Larger weights for minority class



How to evaluate a machine learning model?

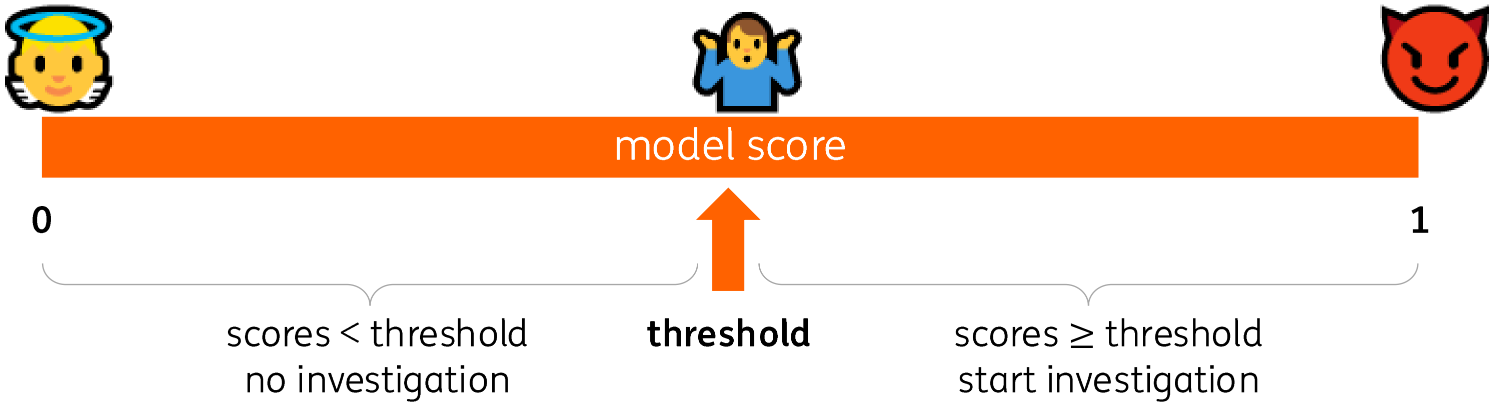
- Accuracy is not a good measure for imbalanced data problems.
  - $accuracy = \frac{nr\ correct\ predictions}{total\ number\ of\ predictions}$
  - E.g.
    - 1% of all companies are risky.
    - The model says "All companies are NOT risky".
    - The model would be wrong only on 1% of the cases: **99% accuracy!**

- We are interested in two metrics for imbalanced data problems:
  - **Precision**: What percentage of companies identified as "risky" are actually "risky"?
    - How correct are the predictions?
    - Low precision = many good companies will be investigated.
  - **Recall**: What percentage of all real "risky" companies did the model find?
    - How complete are the predictions?
    - Low recall means many risky companies are not identified by the model.

Precision-recall trade-off

- Machine learning models output scores between 0 and 1
  - A company with a model score of 0 is unlikely to be a risky company
  - Large model scores indicate large probability of riskyness

- Question: for which score values do we need to investigate a company?



- Where we decide to put a threshold is a trade-off between precision and recall
- Demo!

Generative AI in Banking

Customer Due Dilligence

Transaction Monitoring

Testimony from an anonymous analyst

"

When I joined the PTM (post transaction monitoring) department, the average alert handling per person per day was maybe 2, probably under. As of last summer, that has been upped dramatically, and everybody needs to be able to do 5 (and over, preferably).

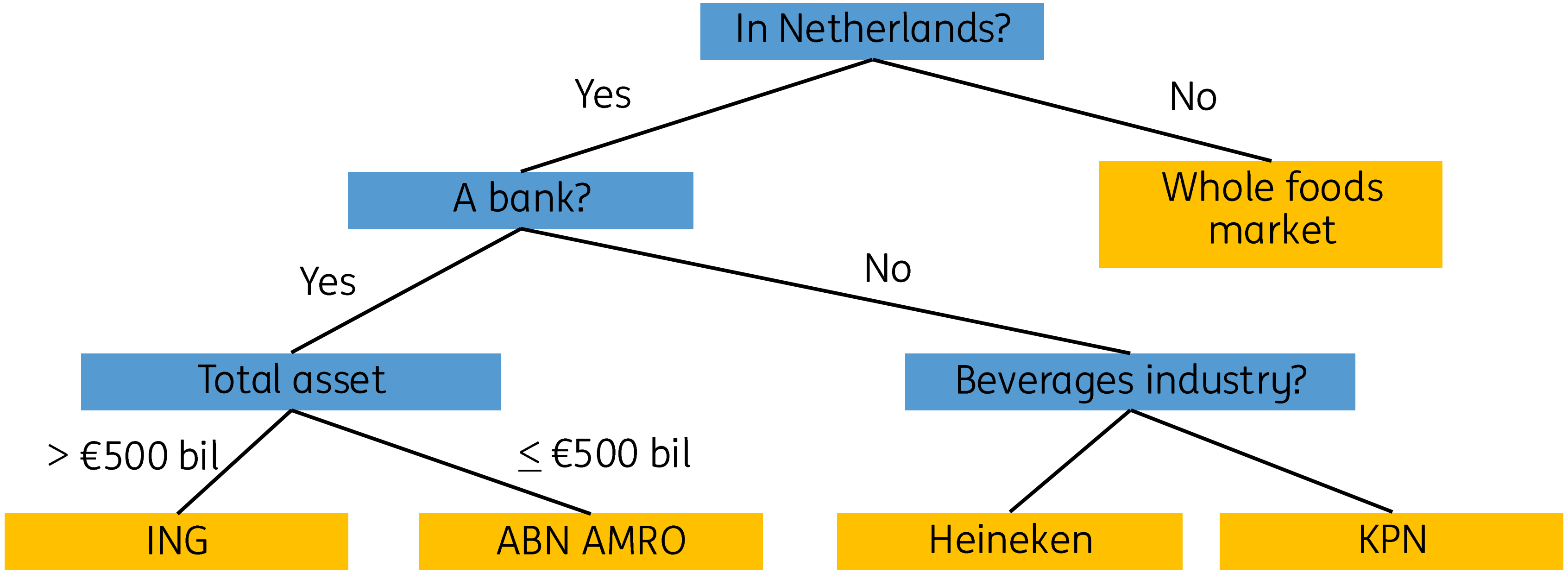
“

Questions

Appendix

Decision trees – a simple decision tree

A flow-cart with decision points

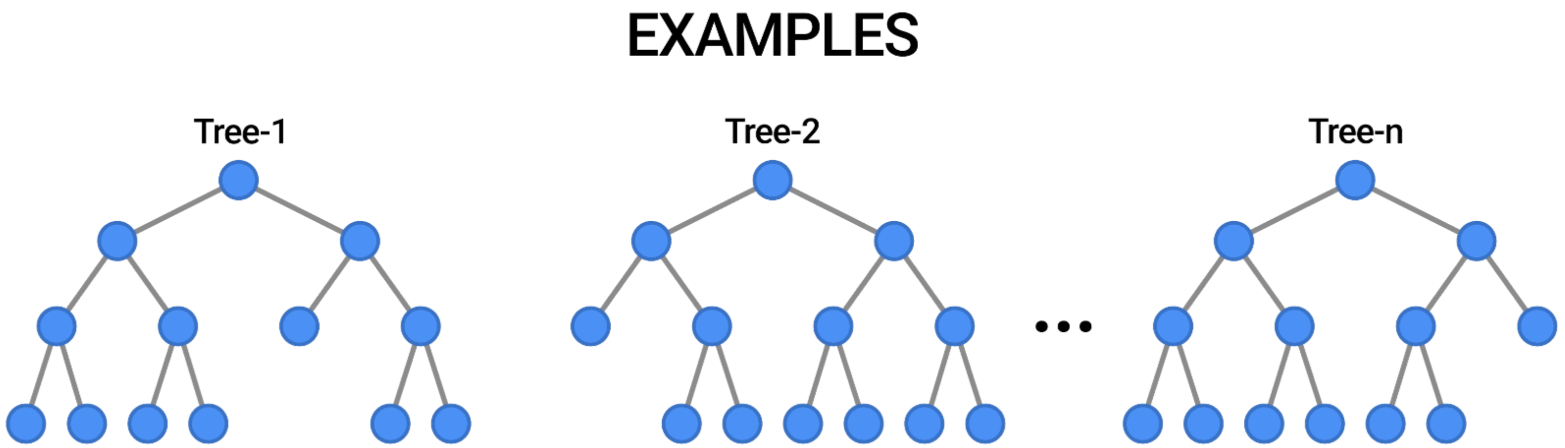


Features
In Netherlands
Banking industry
Beverages industry
Total assets

Decision
ING
ABN AMRO
Heineken
KPN
Whole foods market

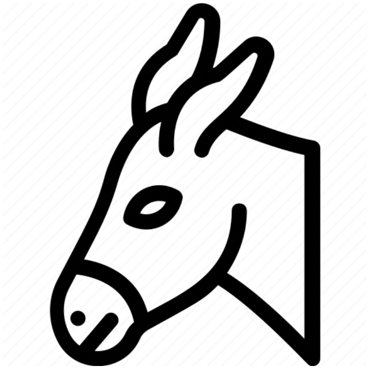
Decision trees – how multiple trees work

Multiple trees are created. Final decision is by majority vote. Strong evidence supporting the enhanced intelligence by a group of trees versus just one tree.



Anti Fraud

Transaction to Mule accounts



Mule accounts are accounts:

- Created by criminals to receive fraudolent money.
- Legitimate accounts of customers that have knowlingly on unknowingly allowed criminals to use their account for illegitimate reasons.

Detecting transactions to Mule accounts using Machine Learning has many challenges:

- It requires large amounts of data
- Data imbalance
- Need to happen real-time, many models would be too slow!

do your thing